

Plasmapheresis

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KEY POINTS

- TPE can be effective in removing large toxins and pathogenic antibodies when combined with immunosuppression.
- TPE is considered first-line treatment in the management of patients with pulmonary hemorrhage in the setting of anti-GBM disease or vasculitis.
- TPE is indicated in patients with anti-GBM disease (without advanced kidney failure), and RPGN (with advanced renal failure).
- The evidence for using TPE in cast nephropathy is controversial, but TPE is indicated in patients with hyperviscosity syndrome.
- The only drug for which TPE is indicated is ticlopidine.
- TPE is indicated in renal transplantation for patients with antibody-mediated rejection, desensitization, and recurrent FSGS posttransplantation.
- TPE is effective in nonrenal diseases such as fulminant Wilson disease, familial hypercholesterolemia, and neurologic diseases, including Guillain-Barré syndrome, demyelinating polyneuropathies, and myasthenia gravis.

CLINICAL RELEVANCE

Therapeutic plasma exchange (TPE) permits removal of high-molecular-weight substances such as antibodies, immune complexes, and toxic proteins. When used in combination with other therapies, TPE has been shown to improve patient outcomes in specific diseases significantly.

Although randomized controlled trials are often lacking, TPE has a role in the treatment of several kidney diseases, including antglomerular antibody disease, rapidly progressive glomerulonephritis, thrombotic microangiopathies, and kidney transplantation. In addition, TPE is first-line therapy for nonrenal diseases such as Guillain-Barré syndrome, demyelinating polyradiculopathy, and familial hypercholesterolemia. The procedure is generally well-tolerated, with few side effects, but usually requires central venous catheter placement.

TPE is an extracorporeal therapeutic procedure in which blood from a patient is separated into plasma and other blood components. The plasma is removed and replaced with another solution. The term “plasmapheresis” refers to the same procedure but does not use a replacement solution. In this chapter, TPE will be used instead of plasmapheresis according to the most recent guidelines of the American Society for Apheresis.¹ This chapter will review the history of plasmapheresis, the major conditions for which it has therapeutic benefit, and technical aspects of the TPE process.

HISTORICAL PERSPECTIVE

The word “apheresis” is derived from the Greek word that means “taking away” or “separation.” It is unclear when the notion of therapeutic removal of blood components first

originated, but it was flourishing even before Hippocrates in the fifth century BC. Bloodletting to remove evil humors was a commonplace medical practice, in part due to a lack of understanding of disease processes and the paucity of effective therapies. By the Middle Ages, surgeons and barbers were specializing in this bloody and often painful practice and, even as late as the 19th century, bloodletting was used for nearly every infectious and malignant malady afflicting patients in the United States and Europe.² The first true therapeutic plasma exchange involving the removal of the bad blood and replacement with a clean solution was performed in 1914 by Abel, Rowntree, and Turner at the John Hopkins Hospital. The procedure was called “vivi-diffusion” and demonstrated the principle that the blood of a living animal could be dialyzed outside the body and then returned to the circulation.³ In 1960, Schwab and Fahey performed the first therapeutic manual plasmapheresis to reduce elevated globulin levels in a patient with Waldenstrom macroglobulinemia.⁴

In the early days, the utility of TPE was based on anecdotal or uncontrolled studies but, in recent years, the number of clinical indications for this procedure has been growing. However, the number of clinical conditions that have been rigorously studied with prospective and randomized controlled trials remains small, and decisions for the implementation of TPE still rest on anecdotal and uncontrolled studies in most cases. A summary of disorders treated with TPE and the evidence supporting its use for each condition can be found in the recent therapeutic apheresis guidelines.¹

GENERAL PRINCIPLES OF THERAPEUTIC PLASMA EXCHANGE

The mechanism(s) for clinical improvement of kidney diseases by TPE depends on the pathophysiology of the underlying disease. TPE should be considered when the pathogenic factor is a high-molecular-weight substance or when the patient is deficient in a plasma component. However, hemodialysis and hemofiltration are more efficient procedures for the removal of small molecules and toxins with large volumes of distribution. The success of TPE in treating any disease characterized by accumulation of toxic proteins or antibodies depends on two variables: (1) the rate of production of the abnormal protein or antibody; and (2) the efficiency of removal with TPE. This balance will determine whether an abnormal component can be removed sufficiently rapidly to provide clinical benefit. This occurs in parallel to the end-organ damage, so the ultimate benefit of the procedure depends on the efficiency of toxin reduction and the rate of end-organ damage. As a result, TPE is rarely used in isolation, but is most often used in combination with immunosuppressive strategies to decrease production and reduce inflammation. In addition to the removal of toxic proteins or replacement of deficient ones (Box 66.1) with plasma exchange, other benefits may include reversal of impaired splenic function to remove immune complexes, removal of fibrinogen, replacement of humoral factors, changes in natural killer cell numbers and activity, decline in B cells, an increase in T cells, an increase in T-suppressor cell function, and a shift from a T helper 2 (Th2)-predominant pattern to a Th1-predominant pattern.⁵

Box 66.1 Pathologic Factors Removed by Plasmapheresis

ADAMTS 13 (metalloproteinase)
Autoantibodies
Complement products
Cryoglobulins
Immune complexes
Lipoproteins
Myeloma proteins
Protein-bound toxins

PLASMAPHERESIS USE IN RENAL DISEASE

The clinical indications of TPE for renal diseases according to category are summarized in Table 66.1.

ANTI-GLOMERULAR BASEMENT MEMBRANE DISEASE

Anti-glomerular basement membrane (anti-GBM) disease is a disorder in which circulating antibodies are directed against the noncollagen (NC)-1 domain of the α_3 chain of type IV collagen, resulting in rapidly progressive glomerulonephritis (RPGN). Goodpasture syndrome is classically defined by the triad of pulmonary hemorrhage, RPGN, and circulating anti-GBM antibodies. A recent study has suggested that environmental factors, including infection, may trigger disease in genetically susceptible individuals. More than 90% of patients have circulating anti-GBM antibodies, and the titer of circulating antibodies correlates with disease activity. Approximately 60% to 70% of patients will have pulmonary disease in addition to RPGN and, rarely, a patient may present with pulmonary hemorrhage and no renal involvement.⁶

Before the use of TPE and immunosuppression, the mortality rate exceeded 90%, with a mean survival time of less than 4 months. The use of TPE was introduced for the treatment of anti-GBM disease in 1975,⁷ and numerous uncontrolled studies and case series published in the last 40 years have suggested the beneficial effect of TPE on overall survival and renal survival. Currently, with the combination of TPE, corticosteroids, and cyclophosphamide, the mortality rate has been reduced to less than 20%. The role of TPE in anti-GBM disease is the rapid removal of the pathogenic antibodies, whereas the cyclophosphamide and corticosteroids are essential to prevent additional antibody synthesis and reduce inflammation. A rapid reduction in anti-GBM antibody titers is necessary in view of the speed of glomerular damage, and this cannot be achieved by drug therapy alone.

Rapid initiation of treatment is critical because recovery of kidney function is much more likely in the early phase of disease, before oliguria develops or dialysis is required. Albumin is the replacement fluid of choice but fresh-frozen plasma (FFP) will be needed in patients with pulmonary hemorrhage or at increased risk for bleeding (after kidney biopsy).⁸ Some of the major studies^{9–16} are summarized in Table 66.2; although none are prospective randomized trials, the use of TPE is now considered standard therapy (category I¹ if not presenting with dialysis dependence; category III if

Table 66.1 Summary of Renal Diseases Treated With Plasmapheresis

Disease	Category ^a
Anti-glomerular basement membrane disease	
Dialysis dependence, No DAH	III
DAH	I
Renal failure; not requiring dialysis	I
Rapidly progressive glomerulonephritis (ANCA)	
Dialysis dependence	I
DAH	I
Renal failure; not requiring dialysis	III
Thrombotic thrombocytopenia purpura	I
Thrombotic microangiopathy, drug associated	I
Ticlopidine	III-IV
All others	I
Thrombotic microangiopathy, complement mediated	III
Factor H autoantibodies	III
Complement gene mutations	
Thrombotic microangiopathy, Shiga toxin-mediated (formerly called hemolytic-uremic syndrome [HUS])	
Renal transplant rejection	
Antibody-mediated rejection (ABO compatible)	I
Antibody-mediated rejection (ABO incompatible)	II
Desensitization, living donor (ABO compatible)	III
Desensitization, deceased donor (ABO compatible)	I
Desensitization, living donor (ABO incompatible)	
Recurrent focal and segmental glomerulosclerosis	
Recurrent in transplanted kidney	I
Steroid resistant, native kidney	III
Cryoglobulinemia	II
Systemic lupus erythematosus (severe)	II
Systemic lupus nephritis	IV
Myeloma cast nephropathy	II
Hyperviscosity in monoclonal gammopathy	I
Nephrogenic systemic fibrosis	III

^aCategory I, standard primary therapy; category II, supportive therapy; category III, when evidence of benefit is unclear; category IV, when there is no current evidence of benefit or for research protocols.

dialysis-dependent and no pulmonary hemorrhage). Levy et al.,¹⁴ in an uncontrolled study, have reported the long-term outcomes in 71 patients with anti-GBM disease. All patients received a standard immunosuppressive regimen of oral prednisolone and oral cyclophosphamide plus TPE. Plasma exchange (50 mL/kg, to a maximum of 4 L) was performed by using a centrifugal cell separator daily for at least 14 days or until anti-GBM antibody was undetectable. Human albumin (5%) with added calcium and potassium was used as replacement fluid, and fresh frozen plasma (FFP) (150–300 mL at the end of the exchange) was used in patients with recent surgery or renal biopsy and those with pulmonary hemorrhage. Overall patient survival was 81% at 1 year of follow-up (95% in those with a creatinine level less than 5.7 mg/dL and 65%

in those who presented with dialysis-dependent renal failure). In patients who presented with a serum creatinine concentration more than 5.7 mg/dL (but did not require immediate dialysis), renal survival was 82% at 1 year and 69% at last follow-up. There was only 8% renal survival in patients presenting with dialysis-dependent kidney failure, and all patients who required immediate dialysis and had 100% crescents on kidney biopsy remained dialysis-dependent. The conclusion was that all patients with anti-GBM antibody disease and severe kidney failure who do not require immediate dialysis should be treated with aggressive immunosuppression and intensive TPE. Because pulmonary hemorrhage is associated with high mortality, TPE should be initiated in patients with anti-GBM disease, regardless of the severity of the kidney failure.

A retrospective review¹⁶ of 221 patients with anti-GBM disease from a single center in China comparing corticosteroids alone and corticosteroids plus cyclophosphamide with corticosteroids plus cyclophosphamide and TPE found overall 72.7% patient survival and 25% renal survival at 1 year. The combination therapy of TPE plus corticosteroids and cyclophosphamide had an overall beneficial effect on patient survival (hazard ratio [HR] for mortality, 0.31 [0.16–0.63]; $P = .001$) and renal survival (HR for renal failure, 0.60 [0.37–0.96]; $P = .032$). In 96 patients with kidney-limited disease (no hemoptysis), the risk factors for end-stage kidney disease (ESKD) were oliguria or anuria (HR, 3.34; 95% confidence interval [CI], 2.03–5.50; $P < .001$), initial serum creatinine (doubling from 1.5 mg/dL; HR, 2.13; 95% CI, 1.65–2.76; $P < .001$), and the percentage of crescents (increased by 20%; HR, 1.83; 95% CI, 1.34–2.48; $P < .001$). Only the combination therapy of TPE plus corticosteroids and cyclophosphamide had a beneficial effect on renal survival (HR for renal failure, 0.41; 95% CI, 0.23–0.73; $P = .002$). Only 2 of 63 patients (3%) with serum creatinine more than 6.8 mg/dL remained dialysis-independent at 1 year (see Table 66.2). To date, there has only been one randomized trial in anti-GBM disease,¹⁷ which compared the addition of TPE to cyclophosphamide and corticosteroids on the clinical course of 17 patients with anti-GBM disease. Only 2 of 8 patients who received TPE became dialysis-dependent compared with 6 of 9 in the immunosuppression-alone group, suggesting a trend toward better outcomes in the TPE group. However, the initial serum creatinine concentration and number of crescents were associated with outcome.

After kidney transplantation, allograft survival rates are comparable between patients with anti-GBM disease and those with ESKD due to other causes, although up to 50% may show linear immunoglobulin G (IgG) staining of the glomerular basement membrane. Kidney transplantation in patients with anti-GBM disease should be delayed at least 12 months after antibody levels have returned to normal. The immunosuppression used to prevent allograft rejection and the sustained disappearance of anti-GBM antibodies are thought to be the main reasons why recurrences in the allograft are rare.^{18,19}

RAPIDLY PROGRESSIVE GLOMERULONEPHRITIS

RPGN is characterized by rapid deterioration in kidney function occurring over a few days to a few weeks. Untreated RPGN usually leads to ESKD. RPGN is characterized by severe

Table 66.2 Renal Recovery in Patients With Anti-Glomerular Basement Membrane Antibody Disease^a

Study (Year)	No. of Patients in Study	Patients With Independent Renal Function at 1 Year		Treatment
		Initial Cr Concentration < 5.7 mg/dL	Initial Cr Concentration ≥ 5.7 mg/dL ^b	
Bouget et al. (1990) ⁹	13	50%	0%	Most patients received PE.
Herody et al. (1993) ¹⁰	29	93%	0%	Most patients received PE.
Merkel et al. (1994) ¹¹	32	64%	3%	25 patients received PE.
Andrews et al. (1995) ¹²	15	NA	7%	All patients had Cr concentration ≥600 mmol/L; only 8 patients received treatment.
Daly et al. (1996) ¹³	40	20%	0%	23 patients received PE.
Levy et al. (2001) ¹⁴	71	94%	15%	All patients received PE, C, and CFM.
Saurina et al. (2003) ¹⁵	32	71%	18%	24 patients received treatment with C, CFM, and PE.
Cui et al. (2011) ¹⁶	176	25% ^c	3%	76 patients C, CFM and PE, 59 PE + C, 41C

^aAccording to initial creatinine concentration.^bOr dialysis-dependent.^cThis study used a 6.8-mg/dL cutoff.

C, Corticosteroids; CFM, cyclophosphamide; Cr, creatinine; PE, plasmapheresis.

inflammation and necrosis of most glomeruli and, frequently, by fibrocellular crescents (crescentic glomerulonephritis, GN). There are three major subgroups of RPGN: (1) anti-GBM disease and Goodpasture syndrome (discussed previously); (2) immune complex-mediated processes where there is immune deposition usually resulting from autoimmune diseases such as systemic lupus erythematosus (SLE), postinfectious processes, mixed cryoglobulinemia, and IgA nephropathy; and (3) pauci-immune diseases that are usually (~80% of patients) associated with antineutrophil cytoplasmic antibodies (ANCA), including granulomatosis with polyangiitis (GPA) or microscopic polyangiitis (MPA). There are limited data for the role of TPE in immune complex-mediated processes (see later).

The rationale for using TPE in pauci-immune ANCA-associated diseases (e.g., GPA, MPA) was initially based on the similarity of the renal pathology of these disorders with anti-GBM disease, and some patients will have both anti-GBM and ANCA-related disease. The first use of TPE for the treatment of RPGN associated with GPA was reported in 1977 when the combination of plasmapheresis, oral prednisolone, and cyclophosphamide was associated with rapid recovery of kidney function in five of nine patients.²⁰ However, several studies throughout the 1990s did not demonstrate an additional benefit for the use of TPE in the treatment of ANCA-associated diseases. For example, the Hammersmith Hospital reported a controlled trial of TPE in focal necrotizing glomerulonephritis on 48 patients randomized to conventional treatment with oral steroids, cyclophosphamide followed by azathioprine, with or without intensive TPE (at least five exchanges in the first 7 days). There was no benefit in patients with moderate or severe kidney disease who were not dialysis-dependent at presentation.²¹ However, this study was the first to suggest that some patients who were dialysis-dependent may be able to discontinue dialysis after treatments that

included TPE (10/17 in the TPE group vs. 3/8 in the conventional group).

The Canadian Apheresis Study Group randomized 32 patients with RPGN to receive intravenous methylprednisolone, followed by oral prednisolone, and azathioprine with or without TPE (10 exchanges in the first 16 days). Again, there was no demonstrable benefit of TPE in non-dialysis-dependent patients; however, 3 of 4 dialysis-dependent patients who received TPE were able to come off dialysis compared with only 2 of 7 patients who did not receive TPE.²²

In one of the largest studies to date, Jayne et al.²³ reported on 137 patients with ANCA-associated systemic vasculitis confirmed by kidney biopsy and serum creatinine concentrations higher than 5.7 mg/dL. Patients were randomized to receive seven TPE procedures ($n = 70$) or 3 g of methylprednisolone in divided doses ($n = 67$). Both groups received oral cyclophosphamide and oral prednisolone as maintenance therapy. Dialysis independence at 3 months was the primary endpoint, and 33/67 (49%) of methylprednisolone-treated patients were alive and independent of dialysis compared with 48/70 (69%) of patients who received TPE ($P = .02$). When compared with methylprednisolone, TPE was associated with a 24% reduction in risk of progression to ESKD at 1 year.

A recent meta-analysis has supported the concept that TPE may reduce dialysis dependence, although there was no benefit on survival. Randomized controlled trials for treating renal vasculitis were included in the analysis (31 trials), and 8 included adjuvant plasma exchange. At 12 months, there was a significant reduction in the provision of dialysis in six trials with 235 patients (relative risk, 0.45; 95% CI, 0.29–0.72); the number needed to treat was 4 to 10.²⁴ In conclusion, although patient numbers are relatively small, and the provision of TPE cannot be blinded, available data in aggregate support a beneficial role for TPE in the

treatment of patients with ANCA-associated vasculitis and severe kidney disease (sCr > 500 μ mol or 5.7 mg/dL) at presentation or relapsing disease. In patients with ANCA- and anti-GBM-associated disease presenting with diffuse pulmonary alveolar hemorrhage, TPE is beneficial for recovery and reduction of risk of progression to dialysis.^{25–27} Therefore, TPE is, at present, the best adjunct to immunosuppressive therapy for patients with advanced kidney disease from these mechanisms of injury. Nevertheless, the use of TPE for less severe kidney disease is still not sufficiently investigated.²⁸

Data on the safety and efficacy of TPE are even more sparse in other forms of vasculitis and in children. In patients with Churg-Strauss syndrome or polyarteritis nodosa, TPE does not appear to be of benefit in addition to cyclophosphamide and steroids.²⁹ In children with ANCA-associated vasculitis, the removal of circulating ANCA through TPE is a short-term measure reserved for severe or refractory disease.³⁰ There is little evidence for or against TPE in other causes of RPGN, although there has been one report of benefit in children with rapidly progressive glomerulonephritis from Henoch-Schönlein purpura (HSP).³¹ A recent study has suggested that TPE may be beneficial in kidney transplant recipients with recurrent HSP nephritis, but there have been no prospective studies or protocols indicating the optimal therapeutic regimen in this group of patients.³²

LUPUS NEPHRITIS

Acute and chronic kidney disease are common and potentially serious complications of SLE. Management of patients with active lupus nephritis (LN) comprises an induction treatment followed by long-term maintenance therapy. Treatment of LN is based on the use of corticosteroids plus cyclophosphamide (IVC) or mycophenolate mofetil³³ in patients with active proliferative disease. The use of biologic agents such as rituximab³⁴ has also been considered as an alternative in the treatment of proliferative LN, but recent data have suggested that rituximab offers no additional benefit to conventional treatment.

The use of TPE for patients with proliferative LN was first reported in the 1970s but it was not until the results of the Lupus Nephritis Collaborative Study Group in 1992 that there was a randomized study to examine the safety and efficacy of TPE systematically. The Lupus Nephritis Collaborative Study Group³⁵ was a large, randomized, controlled multicenter trial comparing a standard therapy regimen of prednisone and cyclophosphamide with a regimen of standard therapy plus TPE in patients with severe LN. In this study, 46 patients were randomized to the standard therapy group and 40 were randomized to the TPE group. Histologic categories included LN types III, IV, and V. TPE was carried out three times per week for 4 weeks, and drug therapy was standardized. The mean follow-up was 136 weeks and, although patients treated with TPE experienced more rapid reduction of antibodies to double-stranded DNA and cryoglobulins, the addition of TPE did not improve clinical outcomes. Kidney failure developed in 8 of 46 patients (17%) in the standard therapy group compared with 10 of 40 patients (25%) in the TPE group, and 6 of 46 (13%) died in the standard therapy group compared with 8 of 40 (20%) in the TPE group. Results were similar in magnitude and direction after an extended follow-up of 277 weeks.

Another small trial has confirmed these findings. Wallace et al.³⁶ randomly assigned 18 patients to receive either 6 months of intravenous cyclophosphamide and prednisone or TPE prior to each infusion of cyclophosphamide. In each group, 2 of 9 patients developed ESKD, and 3 of 9 patients went into renal remission at 24 months. Taken together, these studies suggest that the addition of TPE to conventional treatment for LN does not improve the prognosis of LN, despite more rapid reduction in circulating autoantibodies.

In contrast to the studies of the Lupus Nephritis Collaborative Study Group and Wallace et al., some smaller studies have suggested a benefit of TPE in a select group of patients with aggressive LN. Euler et al.³⁷ used TPE and pulse intravenous cyclophosphamide followed by oral cyclophosphamide and prednisone in an uncontrolled study of 14 patients with severe lupus; all 14 patients responded, and 8 remained off therapy for 5 to 6 years. One patient had a major relapse, and two others had a minor relapse at 2 and 3 years. The main adverse effect was herpes zoster; in addition, 4 of 14 women developed irreversible amenorrhea. Danieli et al.³⁸ compared two groups of patients with proliferative LN at 4 years of follow-up. Group I (12 patients) received synchronized therapy with TPE and cyclophosphamide, whereas group II (16 patients) received intermittent cycles of cyclophosphamide. At the end of the follow-up, the patients who received synchronized therapy had a faster remission than the other group, but renal outcomes were not superior to conventional therapy at long-term follow-up.

Although TPE does not demonstrate clear benefits in severe LN, immunoadsorption using protein A, C1q, or dextran sulfate cellulose columns was shown to be useful for refractory LN with antiphospholipid syndrome in a small case series.³⁹ Li et al.⁴⁰ have shown in a series of 9 patients that additional TPE above and beyond conventional induction therapy may benefit patients with LN and severe thrombotic microangiopathy. Thus, the current literature does not support a benefit for the addition of TPE to immunosuppressive therapy for LN, although it may be considered for select patients with severe disease and no response to conventional treatment. As a result, the American Society for Apheresis considers TPE for LN as category IV (no evidence),¹ but TPE should be considered in patients with catastrophic antiphospholipid antibody syndrome in patients with lupus (category II; accepted second-line therapy).¹

MIXED CRYOGLOBULINEMIA

Cryoglobulinemia refers to the presence of serum proteins that precipitate at temperatures below 37°C and redissolve on rewarming. More than 80% of patients with mixed cryoglobulinemia are infected by hepatitis C virus (HCV), although cryoglobulinemia can also be seen in plasma cell dyscrasias and with vasculitis. The glomerular injury is the consequence of glomerular deposition of immune complexes, and the kidney's manifestations may range from isolated proteinuria to overt nephritic or nephrotic syndrome, with variable progression toward ESKD.^{41,42} There have been no randomized, controlled studies of TPE for cryoglobulinemia; however, the goal of removing pathogenic cryoglobulins is rational, and there have been numerous anecdotal case reports and uncontrolled studies showing that TPE may benefit

Table 66.3 Randomized Controlled Trials of Plasmapheresis in Treatment of Multiple Myeloma

Study (Year)	Renal Biopsy	No. of Patients	Dialysis (%)	PE Sessions	Chemotherapy	Outcome	Results
Zucchelli et al. (1988) ⁴⁹	17	29	82.8	5–7	Melphalan and PDN or VAD	Recovery—renal function	13/15 PE group 2/14 control group
Johnson et al. (1990) ⁵⁰	21	21	57.1	3–12	Melphalan and PDN	Recovery—if dialysis-independent	3/7 PE group; 0/5 control group
Clark et al. (2005) ⁵¹	Not stated	104	29.9	5–7	Melphalan and PDN or VAD	Death or dialysis at 6 mo	33/57 PE group; 27/39 control group

PDN, Prednisone; PE, plasmapheresis; VAD, vincristine, Adriamycin, dexamethasone.

patients with severe active disease manifested by progressive kidney failure, severe or malignant hypertension, purpura, and advanced neuropathy.^{43,44} Uncontrolled studies with more than five patients have shown that TPE induces rapid reduction in the “cryocrit” (the proportion of plasma comprised of cryoglobulin), improved kidney function in 55% to 87% of patients, and improved survival (~25% mortality rate) compared with historical data (~55% mortality rate).⁴⁵ Plasma exchange does not prevent the formation of new cryoglobulins or treat the underlying disease, and rebound production can occur after cessation of TPE without therapy directed at the cryoglobulin-producing B cell clones or treatment of the underlying disease process. The unique characteristics of cryoglobulins have led to modifications of the TPE technique to enhance their removal. Cryofiltration is a modified technique that reduces circulating levels of cryoproteins. In this technique, the plasma is cooled in the extracorporeal circuit to precipitate the proteins, allowing a more efficient removal of the pathogenic proteins.⁴⁶

The treatment of hepatitis C has been revolutionized with the introduction of direct antiviral agents, but few data are available on the safety and efficacy of antiviral therapy with direct-acting antivirals in the treatment of HCV-related cryoglobulinemia. Recently, some trials have shown that patients treated with sofosbuvir-based, direct-acting antiviral therapy experience improvement in kidney function.⁴⁷

KIDNEY FAILURE ASSOCIATED WITH MULTIPLE MYELOMA AND OTHER HEMATOLOGIC DISORDERS

Kidney impairment in patients with multiple myeloma (MM) is defined as an acute decompensation of kidney function that results in a serum creatinine level of 2.0 mg/dL or higher. Kidney biopsy is currently not required for diagnosis, but can help distinguish the various myeloma-related kidney diseases and aid in the treatment plan. Kidney disease is a common finding in MM, affecting 20% to 50% of patients at diagnosis, and 10% to 15% will develop ESKD. Impaired kidney function can be caused by a variety of factors, including precipitation of myeloma light chains in renal tubules (Bence Jones proteins) that can lead to direct tubular toxicity. Other factors frequently implicated in MM-associated kidney failure include hypercalcemia, hyperuricemia, amyloidosis, hyperviscosity, infections, and chemotherapeutic agents.⁴⁸

TPE could be of benefit in preventing tubular damage by the removal of nephrotoxic Bence Jones proteins; these

studies are summarized in Table 66.3. An early study of 29 patients with MM and acute kidney injury (AKI) included 24 patients on dialysis and 5 with serum creatinine concentrations more than 5 mg/dL. Patients were randomly divided into two groups: 15 patients received TPE plus standard therapy and 14 patients had standard therapy alone. In the TPE group, 13 of 15 patients recovered kidney function (defined as serum creatinine <2.5 mg/dL) versus 2 of 14 in the control group.⁴⁹ However, Johnson et al. have reported no difference in patient survival or recovery of kidney function in a study of 21 patients who were randomly assigned to TPE plus chemotherapy or to chemotherapy alone. The mortality at 6 months was 20% in each group, which increased to 60% to 80% at 12 months.⁵⁰ In the largest study to date, 97 patients with MM and AKI were randomly assigned to conventional therapy plus five to seven plasma exchanges of 50 mL/kg of body weight of 5% human serum albumin for 10 days or conventional therapy alone. The primary endpoint (death, dialysis, or estimated glomerular filtration rate [eGFR] <30 mL/min/1.73 m²) occurred in 33 of 58 (56.8%) patients in the TPE group and in 27 of 39 (69.2%) patients in the control group.⁵¹

In aggregate, these studies leave the role of TPE in the management of cast nephropathy unresolved. Questions remain about subgroups of patients who may benefit. In general, these studies have suggested caution in considering TPE for the AKI associated with MM.^{52,53} Novel agents are associated with excellent disease response, and bortezomib is a first-line drug in the treatment of patients with severe acute and chronic kidney disease.⁵⁴

Waldenström macroglobulinemia (WM) is a B cell disorder resulting from the accumulation of clonally related IgM-secreting lymphoplasmacytic cells. The morbidity associated with WM is typically mediated by tissue infiltration by neoplastic cells and by the physicochemical and immunologic properties of the monoclonal IgM. In patients with symptomatic hyperviscosity, cryoglobulinemia, or moderate to severe cytopenias, emphasis should be placed on achieving rapid reduction in the burden of plasma paraproteins. In these circumstances, TPE of two or three sessions will be effective in reducing serum IgM levels by 30% to 60%. Treatment should be initiated as soon as possible, with a regimen including bortezomib, dexamethasone, and rituximab to achieve more rapid disease control.⁵⁵

TPE has been widely used in hematologic and oncologic diseases; however, only the following disorders are considered category I by the American Society for Apheresis: (1)

TTP (discussed later); (2) polycythemia vera with severe erythrocytosis; (3) sickle cell disease (red cell exchange); (4) ABO-incompatible bone marrow transplantation (red cell removal from the marrow; TPE in the recipient to eliminate ABO antibodies is considered category II); (5) hyperviscosity in monoclonal gammopathies; and (6) cutaneous T cell lymphoma (photopheresis).¹

SYNDROMES OF THROMBOTIC MICROANGIOPATHY

The thrombotic microangiopathies (TMAs) are multisystem disorders that can affect children and adults, may be hereditary or acquired, and share a spectrum of abnormalities in numerous organ systems. The hallmark clinical features of this syndrome include microangiopathic hemolytic anemia, thrombocytopenia, and organ (including kidney) injury. The pathologic features are vascular damage manifested by arteriolar and capillary thrombosis. Common causes of this syndrome include toxins from bacteria or drugs, abnormal complement activation that can be genetic or autoantibody-induced, production of procoagulant factors (antiphospholipid antibodies), loss of anticoagulants (defects on ADAMTS13), and severe hypertension.^{56,57} According to the most recent ASFA guidelines, the TMA diseases with class I recommendation for TPE procedures are thrombotic thrombocytopenic purpura (TTP), complement-mediated TMA when it is associated with factor H antibodies, and ticlopidine-associated TMA.¹

TTP, also known as TMA-ADAMTS13 deficiency, is a systemic thrombotic illness affecting mostly small vessels. Presenting clinical features of acquired TTP are weakness, gastrointestinal symptoms, purpura, and transient focal neurologic abnormalities. Clinical presentation is diverse, because some patients have minimal abnormalities, whereas others are critically ill. Diagnostic criteria are the presence of microangiopathic hemolytic anemia and thrombocytopenia without another apparent cause; an ADAMTS13 level of less than 10% of normal activity supports the clinical diagnosis of acquired TTP.^{58,59} Multimers of von Willebrand factor (VWF) normally accumulate on the endothelial cell membrane and are rapidly cleaved into normal-sized multimers by the ADAMTS13 protease. TTP results from the accumulation of unusually large VWF (ULVWF) due to insufficient ADAMTS13 activity. The accumulation of ULVWF multimers results in platelet microthrombi formation and subsequent microangiopathic hemolytic anemia. An inhibitory autoantibody to the ADAMTS13 metalloproteinase has been found at varying titers among a high percentage of patients with the idiopathic form of this disease.^{60–62}

Prior to the introduction of plasma infusion and TPE, the disease rapidly progressed and was almost uniformly fatal (90% mortality). In 1977, it was discovered that infusion of FFP or TPE with FFP replacement was able to reverse the course of disease.^{63,64} Currently, daily TPE is the mainstay treatment by removing the inhibitory autoantibody and repletion ADAMTS13 with FFP. Plasma exchange with FFP is more effective than plasma infusion alone; at 6 months, the remission rate is 78% versus 31%, and the survival rate with these two procedures is 78% versus 50%.⁶⁵ If there is a delay in initiating TPE, large-volume plasma infusions are indicated as initial therapy.⁶⁶

The optimal duration of TPE is not known but, in TTP, TPE is performed daily until the platelet count has risen to near-normal and evidence for hemolysis (schistocytes, lactate dehydrogenase [LDH] elevation) has resolved. There is a wide range of reported exchanges (3–145), with an average of 7 to 16 daily exchanges necessary to induce remission.^{59,67} The ASFA guidelines recommend daily TPE until the platelet count is above 150,000/L for 2 or 3 consecutive days, and the LDH level is close to normal.¹ When present, neurologic symptoms usually improve rapidly, and the serum LDH level tends to improve over the first 1 to 3 days. Improvement in the platelet count may not be seen for several days, however, and improvements in kidney function often take longer. Patients requiring dialysis at presentation may be able to recover enough function to discontinue dialysis, but many patients have residual chronic kidney disease (CKD). When a normal platelet count has been achieved, plasma exchange is gradually tapered by increasing the interval between treatments. Many patients (one-third to one-half) will abruptly develop recurrent thrombocytopenia and increased evidence of hemolysis when daily plasma exchanges are tapered or stopped. Some of these patients may benefit from the addition of prednisone or other immunosuppressive therapy (e.g., cyclosporine, rituximab), although there are sparse data evaluating the safety, efficacy, and effectiveness of these agents.

Although there is no evidence for a beneficial role of TPE in patients with hemolytic-uremic syndrome (HUS), TTP is secondary to most drugs, including cancer chemotherapy and calcineurin inhibitors or when associated with bone marrow transplantation.⁶⁸ There is a single uncontrolled report of 60 patients with ticlopidine-associated TMA, suggesting that TPE enhances survival (50% vs. 24% mortality).⁶⁹ One well-defined cause of HUS (now called TMA–Shiga toxin-mediated) is the syndrome associated with hemorrhagic diarrhea caused by *Escherichia coli* O157:H7. In this disease, the enterotoxin induces colonic vascular injury, leading to systemic absorption and activation of numerous pathways, causing endothelial cell damage over several days. Platelet microthrombi are particularly prominent in the glomerular capillaries and often lead to severe AKI. Supportive care with fluid resuscitation and timely renal replacement therapy are the mainstay of therapy, glucocorticoids are not indicated, and there is no compelling evidence from the literature that TPE benefits patients with this syndrome. However, patients with severe bloody diarrhea or neurologic involvement may respond to TPE. During the 2011 outbreak in Germany, TPE was carried out in 87% patients, yet evidence of benefit was not seen, and improvement in general medical care rather than the frequent and early use of TPE was the potential explanation of the better outcome in comparison to historic controls. A group of patients with more severe disease received TPE plus eculizumab, but survival was not enhanced with addition of this agent.⁷⁰

KIDNEY TRANSPLANTATION

TPE has been used in several different clinical scenarios involving kidney transplantation. These include recurrent primary focal glomerulosclerosis (FSGS) in the transplant, ABO blood group-incompatible transplants, positive T-cell crossmatch, and acute humoral rejection.

RECURRENT FOCAL SEGMENTAL GLOMERULOSCLEROSIS

Primary FSGS is a common cause of end-stage renal failure, and recurrent primary FSGS occurs at a rate of 20% to 30% in kidney transplant recipients. In addition, the risk of relapse is higher (80%–90%) in patients with a prior history of allograft loss due to recurrent FSGS. Additional factors associated with an increased risk of recurrence are rapid progression to ESKD, heavy proteinuria, mesangial hypercellularity, and younger age.

The mechanisms of recurrent FSGS and early detection of proteinuria after kidney transplantation are unclear, but the early reappearance of proteinuria suggests the presence of a nondialyzable circulating factor that alters glomerular permeability. Removal of a circulating factor by immunoadsorption or plasma exchange may account for the remission of the disease in some patients. To date, several circulating factors have been proposed to be the triggers of FSGS, including the soluble urokinase plasminogen activating receptor (suPAR), anti-CD40 autoantibody, and cardiotrophin-like cytokine factor-1 (CLCF1). suPAR represents a soluble (circulating) form of an otherwise membrane-bound, three-domain receptor for urokinase that mediates extracellular proteolysis. The suPAR signaling protein has been shown to be expressed on a variety of cells, including active leukocytes, endothelial cells, podocytes, and immature myeloid cells.^{71–73} In renal transplantation patients, serum concentrations of suPAR were not associated with recurrence, but urine concentrations were (odds ratio [OR], 2.67; 95% CI, 1.23–5.77; $P = .013$).⁷⁴

No consistently effective therapeutic approach has yet emerged. Treatment options include immunosuppressive drugs (e.g., cyclophosphamide, methylprednisolone, high-dose cyclosporine, rituximab) and TPE.⁷⁵ TPE has been used as a first-line treatment for recurrent FSGS, but there are few randomized controlled studies, and most reports have been retrospective reviews of a single center, often without controls.⁷⁶ Nevertheless, TPE has been widely applied in the prevention and treatment of recurrent FSGS recurrence since the initial 1985 report by Zimmerman.⁷⁷ Since then, numerous systematic reviews have shown that TPE promotes partial or complete remission in 70% of children and 63% of adults with recurrence of prior FSGS.⁷⁸ The addition of rituximab has been useful for treating recurrent FSGS in small case series.^{79,80} Rituximab is now widely used as a treatment for recurrent FSGS in cases of incomplete remission or TPE dependence, despite conflicting results, with some studies showing a transient or absent response.

Preemptive treatment of high-risk individuals with TPE in the pretransplantation or perioperative period may alter or even prevent disease recurrence. Ohta et al.⁸¹ have reported the recurrence rate of primary FSGS in patients who received preoperative TPE. TPE was performed in 15 patients and recurrence developed in 5 (33%), whereas 4 of 6 (66%) without preoperative TPE developed recurrence. Gohh et al.⁸² have reported on 10 patients at high risk for FSGS recurrence because of rapid progression to renal failure ($n = 4$) or prior transplantation recurrence of FSGS ($n = 6$). Patients underwent a course of 8 TPE treatments in the perioperative period; 7 patients, including all 4 with first grafts and 3 of 6 with prior recurrence were free of recurrence at follow-up

(238–1258 days). Final serum creatinine concentrations in the 8 patients with functioning kidneys averaged 1.53 mg/dL. Therefore, the use of preoperative and prophylactic postoperative TPE appears promising in high-risk patients, but controlled multicenter trials are lacking. An alternative approach has been to use immunoadsorption columns.⁷⁹ In a recent multicenter French study, the use of early immunoadsorption showed a remission rate of 83%.⁸³

ABO-INCOMPATIBLE KIDNEY TRANSPLANTATION

The ABO blood group antigen system was discovered on red blood cells by Landsteiner in 1901; these antigens are expressed throughout the body and, in the kidney, are found in the distal tubules, collecting tubules, and vascular endothelium of peritubular and glomerular capillaries. The ABO antibodies (isoagglutinins) are produced in the first years of life by sensitization to environmental substances such as food, bacteria, and viruses and are usually of the IgM type. These antibodies against ABO antigens generally prohibit kidney transplantation across ABO barriers and are key mediators of antibody-mediated rejection.

In the early days of kidney transplantation, results with ABO-incompatible (ABOi) transplants were disappointing, but a major breakthrough came with the first large study on ABOi kidney transplantation reported by Alexandre et al.^{84,85} Recipients received two to five TPE sessions, pretransplantation immunosuppression, and splenectomy. Graft survival rates were better in younger patients (89% at 5 years in patients <15 years old compared with 77% in older patients).

In Japan, ABOi renal transplantation flourished in the 1990s, and the outcomes to date have been excellent. In the first report of 441 ABOi kidney transplantations performed at 55 centers across Japan from 1989 to 2001, graft survival was 84% in the first year and 59% at 9 years of follow-up. Allograft survival was not significantly lower than in historical controls who received ABO-compatible living donor organs. For the period of 2001 to 2010, patient and graft survival rates for the 1427 analyzed patients were an excellent 98% and 96% for the first year and 83% and 91% after 9 years, respectively.⁸⁶ The protocol to prepare recipients consisted of four components: (1) extracorporeal immunomodulation to remove AB antibodies before the transplantation; (2) use of immunosuppressive drugs; (3) splenectomy; and (4) anticoagulation. TPE and immunoadsorption were the two techniques performed to remove the AB antibodies, and the goal with either technique was to decrease pretransplantation serum AB titers by 8- to 16-fold. Antibody removal was usually not performed after transplantation.

Splenectomy was more recently replaced by the anti-CD20 antibody rituximab.

The Johns Hopkins group has established a preconditioning protocol of TPE, cytomegalovirus (CMV) hyperimmune globulin (CMVig), and anti-CD20 (rituximab) to allow ABOi kidney transplantation without splenectomy. The treatment protocol requires four to five preoperative TPE sessions to remove anti-A/B antibodies, and each session is followed by the administration of CMVig. After achieving a pretransplant A/B-antibody titer less than 1:16, a single dose of rituximab is given 1 or 2 days prior to transplantation. Thereafter, immunosuppression with tacrolimus and mycophenolate mofetil is initiated, followed by steroids and daclizumab after

transplantation. Postoperative treatment included another three TPE-CMVig sessions on days 1, 3, and 5. The 5-year graft survival rate for the cohort of 60 consecutive patients was 88.7%.^{87,88} Data from the Scientific Registry of Transplant Recipients on the outcomes of 738 ABOi kidney transplantations performed between 1995 and 2010 in the United States have shown a cumulative incidence of graft loss during the first year of 5.9% in ABOi (vs. 2.9% in ABO-compatible recipients). This occurred mainly during the first 2 weeks after surgery due to rejection.⁸⁹

POSITIVE T CELL CROSSMATCH

High sensitization to human leukocyte antigen (HLA) indicates positive T cell crossmatches against multiple potential donors. The degree of sensitization is quantified as the percentage of the donor pool against which the serum of the patient has positive T cell crossmatches—the panel-reactive antibody (PRA) status. Patients with a PRA persistently higher than 50% are generally considered as highly sensitized; primary sensitization results from exposure to foreign HLA antigens via transplantation, transfusion, or pregnancy, although infection and other conditions can also alter sensitization status. Patients with preformed antibodies against HLA antigens have a lower probability of receiving a matched kidney from a deceased or living donor. Furthermore, pre-sensitized recipients enjoy less favorable outcomes after deceased donor kidney transplantation and are at increased risk for hyperacute or acute antibody-mediated rejection and graft loss. Successful transplantation in these patients requires a desensitization protocol of the recipient to a specific donor to reduce the risk of hyperacute rejection and immediate graft loss.^{90,91}

The general approach in HLA antibody reduction protocols involves the use of low-dose IgG and TPE. TPE is used to remove anti-HLA antibodies and is followed by infusion of low doses of IgG during hemodialysis. The rationale is that low-dose IgG will have beneficial immunomodulating effects. Concurrent with TPE initiation, patients are treated with tacrolimus, mycophenolate mofetil, steroids, and antimicrobial prophylaxis. TPE is continued three times weekly until the T cell crossmatch is negative, and transplantation usually takes place within 24 hours. TPE and low-dose IgG are usually repeated several times during the first 2 posttransplantation weeks to remove any rebounding antibody.⁹² TPE-based protocols are usually not suitable for highly sensitized patients awaiting deceased donor transplantation because the availability of suitable organs is unpredictable, and TPE is both difficult and very expensive to continue indefinitely. If TPE is stopped, anti-HLA antibody titers will rebound.⁹³

ACUTE HUMORAL REJECTION

Acute humoral rejection (AHR) is characterized by a severe allograft dysfunction associated with the presence of circulating donor-specific antibodies. The primary aims of therapeutic modalities for AHR are to remove existing antibodies and inhibit their redevelopment. Very poor outcomes are seen with AHR, and treatment with pulse steroids and antilymphocyte therapy is often ineffective. There are few randomized controlled studies using TPE, and most of them were conducted before current high-potency immunosuppression

regimens were available.^{94,95} Removal of the donor-specific antibodies with TPE has been successful when combined with tacrolimus and mofetil mycophenolate.⁹⁶ It is now proposed that the combination of TPE and IVIG may lead to short-term recovery from acute antibody-mediated rejection in more than 80% of cases.⁹⁷ Lefaucheur and colleagues⁹⁸ have compared outcomes of 12 patients with a TPE + intravenous immunoglobulin (IVIg) + rituximab regimen versus 12 patients with IVIg alone, and found better outcomes using TPE + rituximab (91.7% vs. 50% graft survival at 36 months).

Immunoadsorption allows more specific and effective clearance of circulating immunoglobulins, and some authors have proposed that this technique may provide a more effective and selective antibody depletion in few hours. It has been considered as an effective strategy for rapid desensitization in deceased donor transplantation.^{99,100}

OTHER RENAL DISEASES

Dialysis-related amyloidosis is a serious complication of long-term hemodialysis, and it has been considered that accumulation of β_2 -microglobulin in tissues is involved in its pathogenesis. Treatment with direct hemoperfusion column for adsorption of β_2 -microglobulin has been shown to improve symptoms and/or prevent the progression of dialysis-related amyloidosis in most patients.¹⁰¹

Nephrogenic systemic fibrosis (NSF) is a rare but severe systemic disorder that has been rarely observed in patients with acute or chronic kidney disease (more commonly in ESKD), yet it is strongly associated with exposure to select gadolinium-based contrast agents using in magnetic resonance imaging (MRI). Fortunately, the incidence of NSF has declined sharply since 2013. Case reports were published, with moderate treatment success using TPE when symptoms progressed^{102,103}; photopheresis has also been used, with some reported success.

Catastrophic antiphospholipid syndrome (CAPS) is a rapidly progressive and life-threatening disease that results in thromboses in multiple organs in the presence of antiphospholipid antibodies. Rapid-onset thromboses in multiple organs and extensive small and medium vessel involvement in atypical locations are the general characteristics of CAPS. Treatment with anticoagulation, corticosteroids, and TPE or IVIg can be initiated. TPE can remove pathologic antiphospholipid antibodies as well as cytokines, tumor necrosis factor- α (TNF- α), and complement products. Although TPE improves outcomes in patients with CAPS, note that most reported CAPS patients received TPE FFP as the replacement fluid. FFP contains natural anticoagulants (e.g., antithrombin III and protein C), as well as clotting factors, so it is unknown if TPE per se or the FFP replacement provides the benefits to patients with CAPS. There are currently no randomized controlled trials evaluating the safety and efficacy/effectiveness of TPE in CAPS.^{104,105}

THERAPEUTIC PLASMA EXCHANGE AND NONRENAL DISEASE

According to several national registries, TTP, myasthenia gravis, chronic inflammatory demyelinating polyneuropathy, WM, and Guillain-Barré syndrome (GBS) are the most

frequent indications for plasmapheresis, and there are randomized controlled trials indicating benefits for these disorders.¹⁰⁶ There are now nearly 100 rational indications for TPE, and an exhaustive review of the experimental data supporting the different indications for TPE has recently been published by the ASFA.¹ In many clinical settings, it is the nephrologist who is called for initiation of TPE; therefore, it is essential that nephrologists be generally familiar with the literature supporting the use of this procedure in nonrenal diseases.

Plasma exchange is a well-established therapeutic procedure commonly used in many neurologic disorders of autoimmune cause. It is thought that the beneficial effects of TPE occur through the removal of inflammatory mediators, including autoantibodies, complement components, and cytokines. GBS, myasthenia gravis, chronic inflammatory demyelinating polyneuropathy, and demyelinating polyneuropathy with IgG/IgA are considered as indication category I by the ASFA.^{1,107}

GUILLAIN-BARRÉ SYNDROME

GBS has emerged in the 25 years as the most frequently occurring clinical paralytic disorder, with an annual incidence of two cases/100,000 persons. GBS is dangerous, with 10% to 23% of patients requiring mechanical ventilation; up to 1 in 20 patients may die from complications of the disease. In about 60% of cases, GBS develops shortly after an infection, most commonly caused by *Campylobacter jejuni*. A large number of diverse antibodies against different glycolipids, including GM1, GD1a, and GQ1b, have been described.¹⁰⁸ The disorder is characterized by symmetric weakness, paralysis, and distal paresthesias that usually start with the lower limbs, followed by rapid progression to the proximal limbs and trunk over several days. A typical diagnostic feature is the finding of an increased concentration of protein in the absence of a pleocytosis (albumin-cytologic dissociation) in the cerebrospinal fluid (CSF).¹⁰⁹

TPE is a well-established treatment for GBS. Its benefits over and above supportive care have been demonstrated in two large randomized, controlled, nonblinded, multicenter trials.^{110,111} Substantial benefit has been documented for the time to recover the ability to ambulate with assistance, reduction of the proportion of patients who needed assisted mechanical ventilation, a more rapid recovery of motor function, and time to walk with and without assistance. The French Cooperative Group on Plasma Exchange in Guillain-Barre Syndrome¹¹⁰ has addressed the optimal number of TPE sessions for the treatment of GBS and established that two sessions for patients with mild disability and four sessions for patients with moderate and severe disability were the optimal number of sessions. TPE is considered equally efficacious to IVIg therapy; combined treatment of TPE and IVIg does not seem to provide an additional benefit.¹¹¹

CHRONIC INFLAMMATORY DEMYELINATING POLYNEUROPATHY

Chronic inflammatory demyelinating polyneuropathy (CIDP) is a common and potentially treatable disease, with an estimated prevalence of about 1 to 2 cases/100,000 adults. The occurrence of symmetric weakness in both proximal and distal muscles that progressively increases for more than 2

months is the pivotal symptom in the diagnosis of this disease. CIDP is associated with impaired sensation, absent or diminished tendon reflexes, an elevated protein level in CSF, demyelinating nerve conduction studies, and signs of demyelination in nerve biopsy specimens. The presence of autoantibodies against various proteins and glycolipids of the peripheral nerve in samples of serum and CSF from patients with CIDP may provide a rationale for the therapeutic use of TPE. The most widely used treatments for CIDP consist of IVIg, TPE, and corticosteroids. Therapy should be initiated early in the course of the disease to prevent continued demyelination and secondary axonal loss leading to permanent disability. According to published data, there appears to be no difference in efficacy among these three main therapies.^{112,113}

MYASTHENIA GRAVIS

Myasthenia gravis (MG) is an autoimmune-mediated disorder of the neuromuscular junction, clinically characterized by fluctuating muscle weakness and fatigability. The most common variant of the disease is mediated by circulating autoantibodies against the nicotinic acetylcholine receptor (AChR). Mechanisms responsible for the loss of functional AChR that compromise neuromuscular transmission include the degradation of the AChR, complement-mediated lysis of the AChR, and interference with neurotransmitter binding. In subgroups of patients negative for AChR antibody, antibodies against the receptor tyrosine kinase can be detected.¹¹⁴

The treatment of MG includes thymectomy, acetylcholine esterase inhibitors, corticosteroids, immunosuppressive agents, TPE, and IVIg. It is presumed that elimination of circulating AChR antibodies and other humoral factors account for the observed beneficial effects of TPE. TPE is indicated where there is a need for rapid clinical improvement, including myasthenic crisis, impending crisis, preoperative stabilization, and patients in whom long-term control of symptoms is suboptimal with other forms of therapy. Occasional patients require long-term outpatient exchange to achieve adequate control of MG symptoms. Treatment consists of four to six exchanges, each removing 3 to 5 L of plasma, performed daily or every other day. With this TPE regimen, MG symptoms improved within 2 to 3 weeks in 65% of patients but rarely lasted more than 10 weeks without other immunosuppressive therapy. Some patients may have a more prolonged response.^{115–117}

FAMILIAL HYPERCHOLESTEROLEMIA

For familial hypercholesterolemia (FH), the successful use of an apheresis procedure was first described in 1975. Clinical practice recommendations include lipid apheresis as a cornerstone in therapy for lipid lowering in homozygous familial hypercholesterolemia (HoFH) and severe heterozygous familial hypercholesterolemia (HeFH) when traditional lipid-lowering drugs are not sufficiently effective. It is now recommended that low-density lipoprotein (LDL) apheresis should be the treatment of choice for FH homozygotes from the age of 7 years unless their serum cholesterol level can be reduced by more than 50% or decreased to less than 9 mmol/L by drug therapy. It is also indicated for individual patients with heterozygous FH or a family history of premature cardiac death, progressive coronary disease, and where LDL

cholesterol remains higher than 5.0 mmol/L or is decreased less than 40% with maximal drug therapy. Apheresis may also occasionally be indicated in patients with aggressive progressing coronary disease and lipoprotein (a)—Lp(a)—and LDL cholesterol higher than 60 mg/L and 3.2 mmol/L, respectively, despite maximal drug therapy.^{118,119} The use of PCSK9 inhibitors may obviate the role of LDL pheresis in these patients; 11 patients were successfully transitioned to biweekly evolocumab (PMID no. 28926730).

TOXINS AND DRUGS

TPE has also been used to remove toxins, depending on the effective clearance, plasma protein binding, and volume of distribution of the toxic substance. TPE is used to treat mushroom intoxications by *Amanita phalloides* but some reports have suggested that forced diuresis is the treatment of choice.¹²⁰ There is controversy about the beneficial effect of TPE for the treatment of life-threatening intoxications with tricyclic antidepressants, benzodiazepines, quinine, and phenytoin. Other drugs such as L-thyroxine, verapamil, diltiazem, carbamazepine, theophylline, and heavy metals are removed effectively by TPE but the overall change in total body toxin level is usually not clinically significant. Due to the lack of controlled studies, it is difficult to make recommendations for the treatment of poisonings and overdoses.¹²¹

PREGNANCY AND THERAPEUTIC PLASMA EXCHANGE

TPE can be safely performed during pregnancy; introduction of this procedure during pregnancy for diseases requiring that procedure has improved maternal and fetal survival rates. TPE has been safely carried out in myasthenic crisis, Guillain-Barré syndrome, anti-GBM disease, acute fatty liver of pregnancy, and TTP. Until TPE was used for TTP during pregnancy, maternal survival was rare, and the fetal mortality rate approached 80%. In the last 2 decades, a series of reports have revealed the efficacy of plasma exchange, and TTP in pregnancy became a curable disease, with a response rate of about 80% and minimal or no sequelae. LDL apheresis has been tried with varied success in acute hypertriglyceridemia-induced pancreatitis in pregnancy and also as a prophylactic measure for the prevention of pancreatitis in pregnant women with known hypertriglyceridemia.

When maternal plasma contains an alloantibody against a red cell antigen carried by the fetus, hemolytic disease of the fetus and newborn (HDFN) occurs. The maternal IgG crosses the placenta and causes hemolysis of the fetal red cells. This leads to fetal anemia and, when severe enough, fetal death (hydrops fetalis). Usually HDFN is secondary to anti-D (previously termed “Rh disease”), but it can be caused by a variety of red cell alloantibodies—such as anti-K, anti-C, anti-PP1Pk, and anti-E. TPE removes the maternal red cell alloantibody that causes HDFN. Therefore, it is thought that TPE will decrease the maternal antibody titer and, in turn, the amount transferred to the fetus, thereby protecting it from HDFN. Survival in severe cases of HDFN with the use of TPE and/or IVIg prior to intrauterine transfusion is about 70%.

TPE can result in premature delivery because of the removal of hormones crucial in the maintenance of pregnancy. Other complications can occur due to hypovolemic reaction, allergy, transitory cardiac arrhythmias, nausea, and obscured vision. During the exchanges, hypotension must be carefully monitored and corrected and, in the second or third trimester, it is preferable to place the patient on her left side to avoid compression of the inferior vena cava by the gravid uterus.^{56,122–124}

TECHNICAL ASPECTS

The TPE technique involves the withdrawal of venous blood through a needle or central catheter to separate the plasma from blood cells by centrifugation or membrane filtration and reinfuse the cells, plus autologous plasma or another replacement solution. For most conditions, the aim of the procedure is the removal of pathologic autoantibodies or toxins; the initial treatment goal is to exchange 1 to 1.5 times the plasma volume per TPE procedure. This will lower plasma macromolecule levels by 60% to 75%, respectively. A formula that can be used to estimate plasma volume in an adult is as follows¹²⁵:

$$\text{Estimated plasma volume (in L)} = 0.07 \text{ weight (in kg)} \times (1 - \text{hematocrit [Hct]})$$

For removal of components predominantly restricted to the plasma space, the use of higher exchange volumes will require significantly longer procedure times, without additional clinical benefit. The ultimate clinical success of the procedure depends on the abundance of the abnormal protein in plasma and its rate of production. Unless the removal of the protein by TPE is combined with additional therapies (usually immunosuppressive or cytotoxic) to eliminate or reduce the source of the abnormal protein(s), the procedure is unlikely to provide clinical benefit. The time required to suppress abnormal protein production can take several weeks, which is why TPE protocols often require daily apheresis (or near-daily) for prolonged times.

PLASMA SEPARATION TECHNIQUES

The two major modalities to separate the plasma from the blood during a TPE procedure are centrifugation and membrane filtration (Fig. 66.1). The centrifugation method uses centrifugal force to separate whole blood into plasma and cellular fractions according to their density. The centrifugation process can be intermittent or continuous. In intermittent centrifugation, sequential volumes of whole blood are removed and centrifuged; the cellular fraction is returned to the patient, and the process is repeated until the desired volume of plasma is removed. The blood is pumped from the patient at a flow rate of up to 100 mL/min into the processing unit, which consists of a bell-shaped bowl that rotates at high speed. The denser cellular blood components are centrifuged against the lateral walls and the plasma is removed through a central outlet on the top of the bowl. Each cycle removes about 500 to 700 mL of plasma; usually it is necessary to perform the process five or six times to achieve the goal of 2.5 to 4.0 L (1–1.5 plasma volumes) during a session. At the conclusion of each segment, the packed

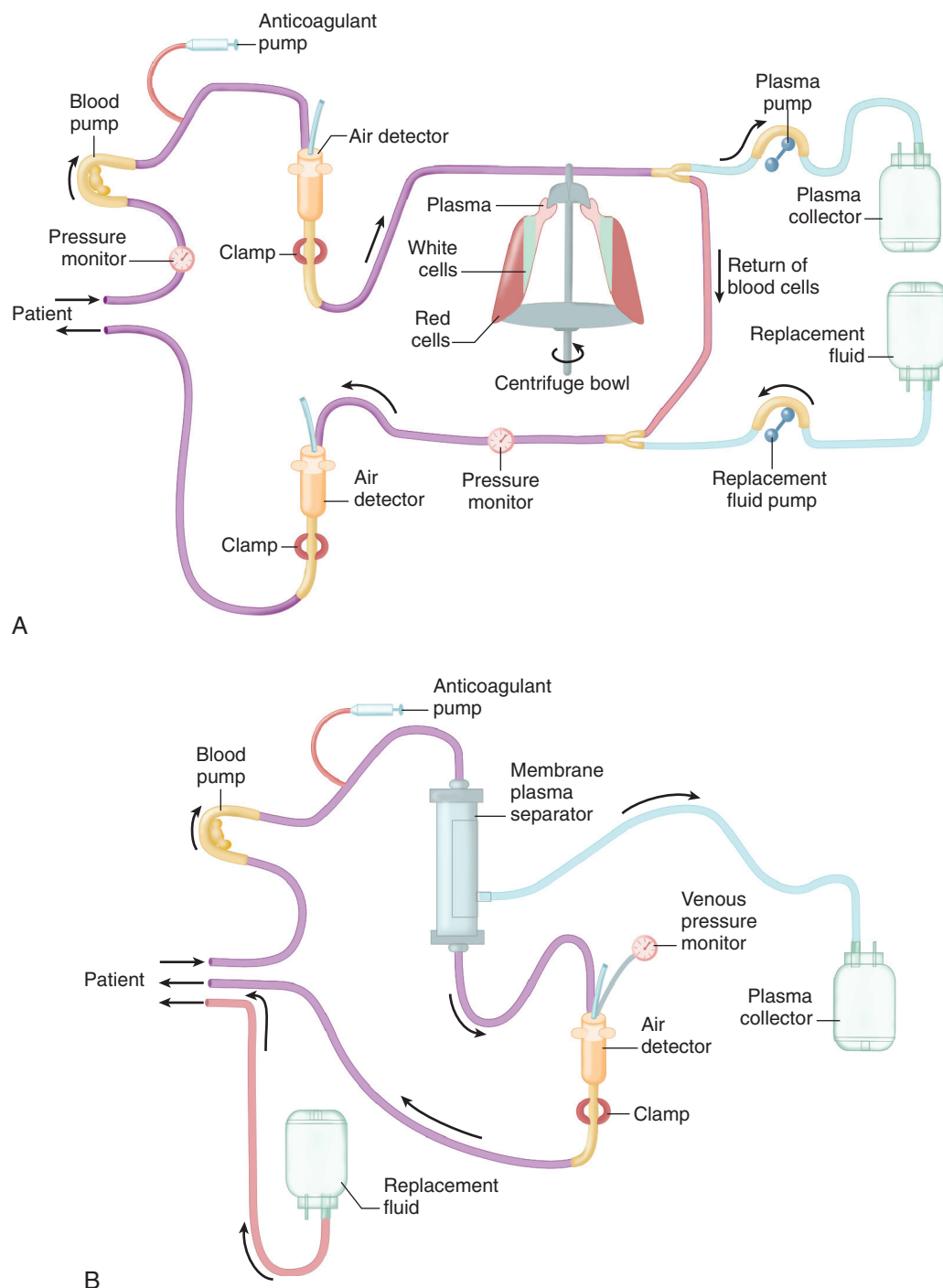


Fig. 66.1 Centrifugal separator (A) and membrane filtration systems (B) for plasma exchange. (A) Blood is pumped into the separator container. As the centrifuge revolves, different blood components are separated into discrete layers, which can be harvested separately. Plasma is pumped out of the centrifuge into a collection chamber. Red blood cells, leukocytes, and platelets are returned to the donor, along with replacement fluid. (B) Blood is pumped into a biocompatible membrane that allows the filtration of plasma while retaining cellular elements. (From Madore F, Lazarus JM, Brady HR: Therapeutic plasma exchange in renal diseases, *J Am Soc Nephrol*. 1996;7:367–386.)

cells are emptied from the bowl and returned to the patient. The advantages of intermittent centrifugation include the relative simplicity of operation, portability of machines, and convenience of a single-needle peripheral venipuncture. The disadvantages include the time (the procedure typically takes >4 hours) and the relatively large extracorporeal volume

removed each time. In the continuous flow centrifugation system, the blood is pumped continuously into a rapidly rotating bowl, where plasma and cells are separated. Plasma is removed at a specified rate, and the cells plus replacement fluid are returned to the patient in a continuous manner. This method is faster and is more suitable for hemodynamically

unstable patients; however, it is more costly and requires two venipunctures or insertion of a dual-lumen central venous catheter.

The membrane filtration technique is based on a synthetic membrane filter composed of different pore sizes. Similar to a hemodialysis filter, the plasmapheresis filter is composed of many hollow fiber tubes made of a membrane material with relatively large pore sizes (0.2–0.6 μm in diameter) and arranged in parallel. Blood is pumped through the hollow fiber tubes, and the large pores are sufficient to allow passage of plasma (proteins and plasma water) while retaining cells within the hollow fiber lumen. The plasma is drained off while the cells are returned to the patient through a typical hemodialysis circuit. This technique can be done using conventional or continuous hemodialysis equipment, with a blood flow rate of 100 (± 20) mL/min and an optimal transmembrane pressure less than 70 mm Hg. Plasma is removed at a rate of 30 to 50 mL/min, and the infusion rate of replacement fluid is adjusted to maintain intravascular volume. Potential disadvantages of membrane filtration include activation of complement and leukocytes by the artificial membrane and the need for a central catheter to obtain adequate blood flow rates. If severe kidney failure is present, and dialysis is required, the membrane filtration method can be done in combination with conventional hemodialysis. Both centrifugation and membrane filtration are safe and efficient TPE techniques; the main differences are the cost and expertise needed to operate.^{126,127}

The double-filtration plasmapheresis (DFPP; also called “cascade filtration”) is another variation of membrane plasmapheresis. In this technique, the plasma that has been separated by the membrane flows again through membranes with different pore diameters and filtration and adsorption characteristics, high-molecular-weight proteins are discarded, and low-molecular-weight substances, including valuable albumin, are returned to the patient. A small amount of substitution fluid such as albumin may be added.¹²⁸

OTHER SEPARATION MODALITIES

In recent years, these basic techniques have been modified and/or coupled to other separation modalities. Cytapheresis is the removal of leukocytes or platelets in hematologic conditions with hyperleukocytosis or thrombocytosis. Cytapheresis can also be performed for sickle cell crisis; in this setting, the goal is the removal of more than 50% of hemoglobin S and replacement with normal allogenic red cells.

When the plasma filtration is carried out above a normal physiologic temperature, the process is called “thermofiltration.” This procedure is performed on patients with severe dyslipidemia, whereas cryofiltration is when the procedure is done with the temperature below normal and is used to remove immunoglobulin and immune complexes. Alternatively, plasma adsorption and immunoabsorption procedures use affinity columns for processing the separated plasma. Adsorption columns such as protein A columns remove IgG antibodies and immune complexes; chemical affinity columns such as dextran sulfate have a negative charge and are used to remove antibodies or other positively charged plasma substances, such as LDLs and very-low-density lipoproteins (VLDLs).¹²⁹

VENOUS ACCESS

Successful implementation of the TPE procedure requires reliable venous access. The clinical scenario, especially the possibility for long-term venous access, and the type of TPE being used are important factors to consider when deciding on peripheral or central venous access. A peripheral vein allows a maximum flow of up to about 50 to 90 mL/min, so a single venous access is adequate for intermittent centrifugation, whereas continuous centrifugation techniques will require two venous access sites. For short-term procedures, this may be adequate but loss of venous access from recurrent intravenous catheters and phlebotomy in chronically ill patients is a major problem. If a long-term TPE is planned (>1–2 weeks), then a central venous catheter is required, preferably one that is tunneled beneath the skin; this generally reduces the likelihood of infection and allows for better catheter performance. When the membrane filtration technique is used, a central venous catheter is necessary to sustain blood flows over 70 mL/min. Central venous access can be achieved through the femoral, internal jugular, or subclavian vein; the femoral vein should be avoided if the treatment of the patient will be ambulatory.¹³⁰ In patients who require lifelong therapy, such as LDL apheresis, either an arteriovenous (AV) fistula or AV graft should be considered.

Central venous catheters have numerous long-term complications, including catheter thrombosis, catheter infection, pneumothorax, central vein thrombosis, and vein stenosis. These issues will not be discussed here; see Chapter 68 regarding the catheters used for maintenance hemodialysis.

ANTICOAGULATION

To prevent activation of the coagulation system within the extracorporeal circuit, the TPE procedure requires anticoagulation. For centrifugation procedures, the acid-citrate-dextrose (ACD) solution (one-ninth the volume of solute/volume of solution), given as a continuous intravenous infusion, is the most frequently used anticoagulant. The infusion rate is adjusted according to the blood flow rate (target ratio ranges from 1:10 to 1:25). When the venous flux and infusion rate of citrate are slow, there is an increased risk for catheter clotting. In this case, heparin (if not contraindicated) can be used alone or in combination with citrate.

For membrane filtration TPE procedures, the use of standard unfractionated heparin is preferred; the required dose of heparin is about twice that needed for hemodialysis because a significant amount of infused heparin is removed along with the plasma. However, heparin may enhance systemic anticoagulation more than expected because of the additional effect of dilution of clotting factors by the non-plasma replacement solutions. The initial loading dose of heparin (40 U/kg) is usually administered intravenously, followed by a continuous infusion (20 U/kg per hour) adjusted to maintain an adequate anticoagulation in the circuit.^{126,127}

For patients who are receiving standard oral anticoagulation—warfarin or one of the newer oral anticoagulants—additional low-dose anticoagulation with regional citrate or heparin should be added to facilitate the treatment and prevent clotting during the procedure. The heparin dose can usually be reduced by at least 50% in this situation.¹³¹ In critically

ill patients with coagulation abnormalities, the use of regional citrate is recommended.¹³² Hirudin and lepirudin (thrombin inhibitors) are effective and relatively safe alternatives for patients with increased risk for thrombosis but have contraindications for heparin administration (e.g., heparin-induced thrombocytopenia, thrombosis).^{133,134}

REPLACEMENT FLUID

The choice of replacement fluids includes 5% albumin, FFP (or other plasma derivatives such as cryosupernatant), crystalloids (e.g., 0.9% saline, Ringer's lactate), and synthetic plasma expanders (e.g., hydroxyethyl starch [HES]).¹³⁵ Albumin is the most commonly used solution in TPE and is generally combined with 0.9% saline on a 50:50 basis. Albumin does not contain calcium or potassium and also lacks coagulation factors and immunoglobulins. It is safe and has never been associated with transmission of hepatitis or HIV viruses.¹³⁶ FFP contains complement and coagulation factors and is the replacement fluid of choice in patients with TTP because the infusion of normal plasma may contribute to the replacement of the deficient plasma factor ADAMTS13 (discussed previously). Plasma may also be preferable in patients at risk of bleeding (e.g., those with liver disease, disseminated intravascular coagulation, after a renal biopsy) or those requiring intensive therapy (e.g., daily exchanges for several weeks) because frequent replacements with albumin solution will eventually result in post-TPE coagulopathy and a net loss of immunoglobulins. The disadvantages of using FFP include the risk of viral disease transmission and citrate overload.

Colloidal starch has been used for a replacement solution. It is well tolerated, drug interaction is minimal, the potential for disease transmission is absent and it is cost-effective when compared with human albumin.¹³⁷ HES is a polysaccharide colloid used as an agent for plasma volume expansion and for enhancement of granulocyte yields during leukapheresis. Its pharmacology and safety have been well studied and, when used in moderate amounts in humans and animals, HES produces relatively minor changes in coagulation tests, and overt bleeding rarely occurs. The exact mechanisms of these changes are unclear, but have been ascribed, at least in part, to hemodilution. In TPE procedures, the side effects of HES on coagulation factors and tests are comparable to those of other replacement fluids. HES can be safely used in the first two sessions of therapy in patients with an albumin level above 30 g/dL to perform short-term TPE¹³⁸; longer HES exposure (130 L within 20 months) can lead to symptomatic effects (e.g., sensory polyneuropathy, weight loss) and diffuse tissue infiltration with HES-laden foamy macrophages.¹³⁹ Excessive HES exposure in patients with impaired renal function can result in acquired lysosomal storage disease, so HES should be avoided in chronic TPE procedures. Contraindications for using starches in TPE are congestive heart failure, renal or liver failure, coagulopathy, hyperviscosity, allergic reactions to starches, pregnancy, breastfeeding, and pediatric patients.¹⁴⁰

COMPLICATIONS

TPE is a well-tolerated and relatively safe procedure and, although adverse events are common, death is rare, occurring

Table 66.4 Complications of Plasmapheresis

Modality	Complications
Vascular access	Hematomas Pneumothorax Catheter infections
Replacement fluids	Anaphylactoid reactions to fresh-frozen plasma Coagulopathies Transmission of viral infections Hypocalcemia Hypokalemia
Other modalities	Hypotension Dyspnea Thrombocytopenia Removal of erythropoietin and drugs bound to plasma proteins

in less than 0.1% of all procedures.¹⁴¹ There are fewer adverse reactions when albumin is administered as volume replacement than when using FFP (1.4% vs. 20%). The high-risk criteria for adverse reactions associated with TPE are unstable vital signs, hypotension, active bleeding, severe bronchoconstriction, severe anemia, pregnancy, and conditions requiring continuous nursing support.¹⁴² Table 66.4 summarizes the most common complications related to the procedure.

The Swedish registry of therapeutic apheresis has reported more than 14,000 procedures from 1996 to 1999; adverse events occurred in 4.2% of procedures, and no fatalities were reported. Of all the apheresis procedures, 1% had to be interrupted due to an adverse event. The most common adverse effects reported were paresthesias (0.52%) hypotension (0.5%), urticaria (0.34%), shivering, and nausea. These adverse events were most frequent in patients with Goodpasture syndrome (12.5%), TTP-HUS (10.5%), and Guillain-Barré syndrome (11.0%).¹⁴³ Kiproff and associates,¹⁴⁴ in another report of 17,940 procedures performed on 3583 patients, had an occurrence rate of adverse events in 3.9% of all procedures. The following adverse reactions were documented: reactions related to citrate toxicity (3%), vasovagal reactions and hypotension (0.5%), vascular access-related complications (0.15%), reactions related to FFP (0.12%), hepatitis B from FFP (0.06%), arrhythmias (0.01%), hemolysis due to inappropriate dilution of 25% albumin (0.01%), and one death (from underlying disease) during a TPE procedure (0.006%). No significant bleeding complications were observed, and patients receiving FFP had significantly higher rates of adverse reactions than patients receiving other exchange fluids.

One of the most frequent complications of TPE is the hypocalcemia related with citrate infusion as anticoagulant for the extracorporeal system or in the FFP administered as a replacement fluid. Citrate binds to free calcium to form soluble calcium citrate, thereby lowering the free but not the total serum calcium concentration. Hypocalcemia may manifest with perioral and distal extremity paresthesias. Symptoms can be prevented and reduced by administration of intravenous or oral calcium if the TPE therapy will last longer than 1 hour. The administration of oral calcium carbonate or addition of calcium gluconate to the return fluid are

useful maneuvers to prevent hypocalcemia.¹⁴⁵ The incidence of hypocalcemic symptoms is lowered with the prophylactic administration of calcium; without calcium prophylaxis, the incidence of symptoms was 9.1% (6 in 66 treatments), whereas with calcium prophylaxis the incidence was reduced to 1% (6 in 633 treatments). Marques and coworkers¹⁴⁶ have reported an incidence of hypocalcemia of 3% when calcium gluconate is infused in 5% albumin.

Another complication of citrate administration is the development of metabolic alkalosis, but serum concentrations of bicarbonate higher than 35 mEq/L are rarely seen. Risk factors more frequently associated with this complication are use of FFP and low GFR (e.g., patients with TTP); this is because the excess of citrate generates bicarbonate, the excretion of which is limited by impaired kidney function. Replacement regimens using saline and albumin solutions can result in a 25% reduction in the plasma potassium concentration in the postapheresis period, which can be minimized by adding 4 mEq potassium/L to the replacement solution. Hypokalemia is also a consequence of metabolic alkalosis.¹⁴⁶

TPE can lead to a reduction in blood pressure that is usually due to a decrease in intravascular volume. Because the volume of extracorporeal whole blood is higher with intermittent centrifugation techniques, hypotension episodes are more common than with continuous modalities. Hypotension can also occur in response to complement-mediated reactions to the membrane filter or sensitivity to the ethylene oxide that is used to sterilize the membrane. Intravenous administration of FFP can rarely result in anaphylactoid reactions, themselves rarely resulting in death. FFP reactions are most often characterized by fever, rigors, urticaria, wheezing, and hypotension.

The development of dyspnea suggests the presence of pulmonary edema due to fluid overload; noncardiogenic edema can rarely occur as a component of anaphylactoid reactions. Rarely, massive pulmonary emboli can develop if the blood components that are reinfused are not adequately anticoagulated.

Plasma exchange with albumin replacement produces a predictable decrease in clotting factors that may predispose to bleeding (Table 66.5). A single plasma volume exchange increases the prothrombin time by 30% and the partial

thromboplastin time by 100%; these changes return toward normal within several hours but, with repeated TPE sessions, these abnormalities can persist. In reported studies, the most significant change is on the fibrinogen. Keller and colleagues¹⁴⁷ have reported that fibrinogen levels decrease to 25% of pre-TPE levels and recover to baseline after 2 to 3 days. Therefore, 1 L or more of FFP (3–4 U/L) should be substituted as the replacement fluid each week or sooner in patients at risk for bleeding. Thrombocytopenia is also a consequence of plasma removal, with larger volumes removed associated with greater platelet loss; the mean platelet reduction following a TPE procedure ranges from 9.4% to 52.6%. Clinical bleeding associated with TPE is rarely reported and, when TPE-related hemorrhage is present, it is more likely a consequence of thrombocytopenia or inadequate heparin neutralization.^{148,149}

Removal of immunoglobulins and complement can result in immunodeficiency, which can exacerbate immune effects of other therapeutic agents (e.g., corticosteroids, cyclophosphamide, rituximab). However, in a randomized controlled trial of TPE in patients with lupus nephritis, TTP, or multiple myeloma, patients receiving TPE were not more prone to infection than the other patients.¹⁵⁰ Nevertheless, repeated apheresis treatments with albumin replacement will deplete the patient's reserve of immunoglobulins for several weeks. If an infection occurs, a single infusion of IVIg (100–400 mg/kg) will restore the plasma immunoglobulin concentration toward normal. Although estimates for the risk of viral transmission by the use of FFP are low, the large volumes from multiple donors increase the risk in patients receiving long-term TPE therapy. Use of large-volume plasma units collected from a single donor and the use of hepatitis B vaccine may reduce the risk of the patient acquiring a virally transmitted infection.

Substantial drug removal by TPE occurs for those drugs that are highly protein-bound and therefore primarily limited to the vascular space. Among drugs used to treat renal diseases, prednisone is not substantially removed, whereas cyclophosphamide and azathioprine are removed to some extent. This potential problem can be circumvented by administering the drug after a plasma exchange treatment.

Flushing, hypotension, abdominal cramping, and other gastrointestinal symptoms have been reported during TPE

Table 66.5 Decrease of Clotting Factors Following Plasmapheresis

Factor	Decrease From Baseline (%)	Factor Level	
		24 Hours After Plasmapheresis	48–96 Hours After Plasmapheresis
V	50–71	RTB	RTB
VII	69–82	62	RTB
VIII	50–82	62, RTB	RTB
IX	26–55	RTB	RTB
X	67–84	RTB	RTB
XI	50–66		RTB
XII	66		RTB
Antithrombin III	58–84	70, RTB	82%, RTB
Fibrinogen	50–78	60	63%, RTB

RTB, Return to baseline.

in patients receiving angiotensin-converting enzyme (ACE) inhibitors. In one report of 299 consecutive patients undergoing TPE, these atypical symptoms occurred in all 14 patients receiving an ACE inhibitor versus only 7% of those not treated with this medication.¹⁵¹ The administration of an ACE inhibitor may prolong the half-life of bradykinin, thus permitting the attainment of a clinically significant concentration in the plasma, so it is recommended to withhold this type of drug for 24 hours prior to TPE.¹⁵²

In conclusion, TPE is a relatively safe procedure, with most of complications being mild and reversible. However, moderate to severe reactions, even death, can occur, especially in patients with severe primary diseases.

CONCLUSION

The use of TPE to treat a variety of conditions has grown significantly in recent years. In some cases, the rationale and benefit are supported by clinical studies, but in many cases the benefits have not been well established. Nevertheless, the concept of removing plasma containing pathogenic antibodies is now well established. Additional studies are needed to determine the potential benefits for TPE in these other conditions.



Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

1. Which of the following mechanisms does not explain how TPE can improve clinical outcomes in patients with renal disease?
- Efficient removal of antigen-antibody complexes
 - Removal of protein-bound toxins
 - Increased activity of the reticuloendothelial system
 - Indicated when the pathogenic factor is a low-molecular-weight substance
 - Immunoadsorption

Answer: d

Rationale: Dialysis would be more effective for low-molecular-weight substances.

2. Which of the following statements involving the benefits of TPE in addition to immunosuppressive therapy in the treatment of anti-GBM is not true?
- Pulmonary hemorrhage is an indication for plasmapheresis.
 - In patients with severe renal failure, plasmapheresis is contraindicated.
 - Plasmapheresis induces a rapid removal of the pathogenic antibodies.
 - Plasmapheresis is considered as standard therapy for this disease.
 - Plasmapheresis has been associated with an increase in renal survival.

Answer: b

Rationale: Although studies show very low rates of renal recovery when patients present with advanced renal failure or dialysis dependence, there is no contraindication to trying it in select patients.

3. Which of the following statements concerning the use of TPE in the treatment of patients with multiple myeloma and cast nephropathy is true?
- The serum creatinine level is an indication for plasmapheresis in this population.
 - Patients with Bence Jones proteinuria benefit more from this procedure.

- High-flux hemodialysis interferes with plasmapheresis effectiveness.
- There are no randomized controlled trials of patients with cast nephropathy showing benefit of treatment with TPE.

Answer: d

Rationale: The few randomized controlled trials available did not require renal biopsy for inclusion and are likely a mixture of causes.

4. Which of the following disorders is not associated with benefit of TPE?
- Guillain-Barré syndrome
 - IgA nephropathy with crescents on biopsy
 - Myasthenia gravis
 - Chronic inflammatory demyelinating polyneuropathy
 - Granulomatosis polyangiitis (GPA) with advanced renal failure

Answer: b

Rationale: There is no evidence for benefit in crescentic IgA nephropathy. All the other conditions are category I.

5. For which of the following drug-associated thrombotic microangiopathies is there evidence of benefit for TPE?
- Mitomycin c
 - Tacrolimus
 - Ticlopidine
 - Clopidogrel
 - Quinine

Answer: c

Rationale: Only ticlopidine has category I evidence supporting the use of TPE.